Ruth Neville

Population Geographer | **Computational Social Scientist**
Postdoctoral Research Fellow, UCL CASA

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PROFILE

Population geographer and computational social scientist specialising in international student mobility, migration systems, demographic forecasting, and population change under disruption. My research examines how educational and migration pathways are shaped by policy, demographic change, institutional relationships, and shocks, and how these processes redistribute people, skills, and opportunities across places. My emerging research agenda examines international student mobility as both an educational trajectory and an upstream component of high-skilled migration and human-capital formation.

Methods: Spatial interaction modelling; longitudinal and spatial analysis; difference-in-differences and causal inference; demographic forecasting; machine learning; network analysis; clustering; data linkage; data visualisation; reproducible research workflows

Programming: R (advanced: `tidyverse`, `sf`, `ggplot2`, `tmap`, R Markdown, Quarto); Python (intermediate: `pandas`, `scikit-learn`, `matplotlib`); QGIS; LaTeX; Git/GitHub

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow in Computational Social Science 2025–present
Centre for Advanced Spatial Analysis (CASA), University College London

- Conduct research using large-scale linked administrative and geospatial datasets, including a national database of approximately 25 million residential properties.
- Develop reproducible workflows for data linkage, harmonisation, spatial analysis, modelling, and visualisation.
- Contribute to collaborative research on population, housing, spatial inequality, and urban change; teach GIS and supervise Master's research.

UCL Public Policy Fellow 2026
London Borough of Newham / University College London

- Developed demographic and spatial evidence on population change, housing, adult social care, and neighbourhood inequalities for local government and health partners.
- Produced small-area indicators, projections, maps, and analytical briefings for policy-facing decision-making.

Research Assistant — Digital Twin of Mass Transit 2024–2025
University of Liverpool

Supported interdisciplinary research on spatial data and urban mobility and coordinated an international UK–South Korea research workshop on data-driven transport planning.

Research Assistant — RECAST / UN Policy Collaboration 2023
University of Liverpool

Contributed to quantitative human-mobility research, data analysis, and international research engagement with ECLAC.

Research Assistant — Mobility Patterns of Ukrainians 2022
University of Liverpool

Analysed large-scale Meta digital-trace data alongside traditional geographic data to examine refugee mobility following the Russian invasion of Ukraine.

Data Analyst 2018–2020
University of Manchester

VISITING RESEARCH POSITIONS

Max Planck Institute for Demographic Research (MPIDR) 2024
Visiting Researcher, Department of Digital and Computational Demography

- Spent approximately five months at MPIDR conducting research on international student mobility and migration.
- Developed international student mobility forecasting research with Emilio Zagheni, subsequently published in *Nature Communications*.
- Participated in research activities on international migration and scholarly and high-skilled mobility.

United Nations Economic Commission for Latin America and the Caribbean (ECLAC) 2023
Visiting Researcher, Santiago, Chile

Contributed to policy-oriented human-mobility research and co-organised research training and stakeholder engagement activities.

EDUCATION

PhD, Geographic Data Science 2020–2025
University of Liverpool

Drivers, Disruptions, and Future Trajectories: Understanding International Student Mobility to the UK

Combined spatial interaction modelling, difference-in-differences, machine learning, and demographic forecasting to examine the structural drivers, disruption, and future trajectories of international student mobility.

Supervisors: Prof. Alex Singleton and Prof. Francisco Rowe

MSc, Geographic Data Science 2020–2022
University of Liverpool

MA, Political Science (Research Route) 2017–2018
University of Manchester

BA (Hons), Philosophy, Politics and Economics — First Class 2013–2016
Lancaster University

SELECTED PUBLICATIONS

International Student Mobility, Education, and Migration

Neville, R., Rowe, F., & Zagheni, E. (2026). Machine learning forecasts suggest a concentration paradox in international student mobility to the United Kingdom. *Nature Communications*, 17, 9604.

Neville, R., Anastasiadou, A. (2025). Gender dynamics in international student mobility: The case of the United Kingdom. *Migration Policy Practice*.

Cabrera, C., & **Neville, R.** (2025). Widely used but barely trusted: understanding student perceptions on the use of generative AI in higher education. *Perspectives: Policy and Practice in Higher Education*.

Neville, R., Rowe, F., & Singleton, A. (2024). Unravelling the Brexit–COVID-19 nexus: Assessing the decline in EU student applications. *Data & Policy*. DOI.

Neville, R., Rowe, F., & Singleton, A. (2024). Nonlinear relationships between human development and international student mobility: The prominent role of employment prospects and cultural and linguistic ties. *Population, Space and Place*. DOI.

Population Mobility and Computational Demography

Cabrera, C., González-Leonardo, M., Nasuto, A., **Neville, R.**, & Rowe, F. (2026). Sustained changes to urban mobility after COVID-19 amplified socio-economic inequalities in Latin America. ***Nature Communications* — accepted for publication.**

González-Leonardo, M., Cabrera, C., **Neville, R.**, Nasuto, A., & Rowe, F. (2025). Beyond the immediate impacts of COVID-19 on internal population movements in Mexico: Facebook data reveal urban decay and slow recovery. *Demography*.

González-Leonardo, M., **Neville, R.**, Gil-Clavel, S., & Rowe, F. (2024). Where have the Ukrainian refugees gone? Integrating digital and traditional geographic data. *Population, Space and Place*.

Housing and Urban Change

Harris, E., Hubbard, P., **Neville, R.**, & Wilkinson, E. (2025). Implementing the nationally described space standard: regulating housing in a devolved planning system. *Journal of Social Welfare and Family Law*, 47(4), 442–463.

SELECTED TALKS AND CONFERENCES

Keynote and Invited Talks

Keynote — RSAI-BIS Annual Conference, Belfast (May 2026)
"Forecasting International Student Mobility in an Era of Uncertainty: From Patterns to Prospects"

BEGIN Seminar, University of St Andrews (2026); University of Bristol (2025); CASA Seminar Series, UCL (2025)
"Drivers, Disruptions and Future Trajectories: Understanding International Student Mobility to the UK"

TU Dortmund — RAM Colloquium and Talk-toranden Series (2024)
"Forecasting International Student Mobility in an Era of Uncertainty: From Patterns to Prospects"

Selected Conference Presentations

2026 BeNeLux Geography, Leuven (Oral); European Population Conference, Bologna (Oral)

2025 International Forum on Migration Statistics, Malmö (Oral); Population Association of America, Washington, DC (Poster); British Society for Population Studies, Swansea (Oral)

2024 Population Association of America, Columbus, OH (Oral and Poster); British Society for Population Studies, Bath (Oral); Conference on Complex Systems, Exeter (Oral); Population Geographies Conference, Belfast (Oral); Scholarly Migration and Mobility Symposium, Max Planck Institute for Demographic Research (Oral)

2023 RGS–IBG Annual International Conference, London (Oral); Symposium on the Relevance of Applied Population Geography, Lesbos (Oral); British Society for Population Studies, Keele (Oral); RSAI-BIS Annual Meeting, Newcastle (Oral); Population Association of America, New Orleans, LA (Poster)

2022 RSAI-BIS Early Career Colloquium (Oral); COST Action European Network on International Student Mobility, Tartu (Oral); GISRUK, Liverpool (Oral)

GRANTS AND AWARDS

- **UCL Public Policy Fellowship associated project funding** — £5,000 (2026)
- **British Academy Early Career Researcher Network Development Fund** — 2026
- **UCL Careers Extra Internship Grant** — £1,500 (2026)
- **Royal Geographical Society QMRG Annual Lecture Funding** — £600 (2026)
- Finalist, University of Liverpool Three Minute Thesis Competition (2024)
- Best Presentation, RSAI-BIS Early Career Colloquium (2022)

TEACHING AND SUPERVISION

Lecturer / GIS Teaching 2025–present
University College London (CASA)

Master's-level teaching in GIS, spatial analysis, regression, reproducible spatial workflows, and R; supervision of Master's dissertation research.

Guest Lecturer and Practical Instructor 2026
University of Oxford

Designed and delivered teaching in spatial analysis for public health, including spatial clustering, autocorrelation, and regression in R.

Teaching Assistant 2022–2025
University of Liverpool

Teaching across Geographic Information Systems, Geographic Data Science, and Social and Spatial Inequalities; undergraduate and postgraduate supervision and assessment.

Associate Fellow, Advance HE (AFHEA)

ACADEMIC LEADERSHIP AND SERVICE

Web and Social Media Officer 2024–present
Quantitative Methods Research Group, Royal Geographical Society

Postgraduate Research Representative 2022–2024
Quantitative Methods Research Group, Royal Geographical Society

Co-Editor, Special Section — *The Geographical Journal* 2025–present
"Mobilising Spatial Data for Spatial Populations"

British Society for Population Studies, 2025–2026: Organiser, chair, and presenter for Novel Data and Methods sessions.

Royal Geographical Society Annual Conference, 2023–2026: Session organiser, chair, and presenter.

QMRG–*Geographical Journal* Guest Lecture, RGS 2026: Organiser; secured £600 in Royal Geographical Society funding and coordinated the associated networking event.

Peer Reviewer: *Population, Space and Place*; *Transactions of the Institute of British Geographers*; *REGION*; *Eastern Economic Journal*.

REFEREES

Prof. Francisco Rowe
Professor in Population Data Science and Lead of the Geographic Data Science Lab (GDSL), University of Liverpool
F.Rowe-Gonzalez@liverpool.ac.uk

Prof. Jon Reades
Professor of Geographic Data Science and Head of Department, UCL CASA
j.reades@ucl.ac.uk